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Topic: Homework Week 14 - Date Display GUI 2 Reflection

The Date Display GUI 2 application has been successfully redesigned using JavaFX, achieving all standard functional requirements. The core functionalities include dynamic real-time date formatting and precise day, month, and year calculations with comprehensive input boundary validation.

Architecturally, the application strictly adheres to the Model-View-Controller (MVC) design pattern to ensure scalability and proper separation of concerns. The DateModel encapsulates data via JavaFX properties, the DateView defines the declarative hierarchy using FXML, and the DateController manages application events and state mediation. Compared to the previous legacy AWT/Swing implementation, which relied heavily on imperative, hard-to-maintain event listeners and manual UI repaint triggers, this JavaFX implementation vastly improves code modularity and responsiveness.

A key enhancement is the utilization of JavaFX properties and reactive framework mechanics. The `textProperty()` of the current display label is dynamically bound using a unified custom `StringBinding` that reacts implicitly to modifications in both the selected date and the selected date format `ComboBox`. Furthermore, the input lifecycle is monitored via an analytical text change listener. It updates a `BooleanProperty`, which is bound directly to the `disableProperty()` of the Apply button. This guarantees that the control interface is automatically and gracefully locked whenever invalid parameters are input, mitigating potential application crashes. Ultimately, the transition to JavaFX eliminates tightly coupled design liabilities, resulting in a cleaner, more robust, and highly maintainable architecture.